ADVANCED SQL

DATE FUNCTIONS.

Exercise 2.2 Practical SQL worksheet.

Question 1: Add 6 months to each employee's hire date using DATEADD()

Table 1: Employees

Emp-id	name	hire_date
1	Alice	2020-01-15
2	Bob	2021-06-10
3	Charlie	2023-03-22

```
SELECT emp-id,
       name,
       hire_date,

  DATEADD(month, 6, hire_data) AS
  hire_plus_6_months
  FROM Employee;
```

Output Table

Emp-id	name	hire_data	hire_Plus_6_month
1	Alice	2020-01-15 00:00:00	2020-07-15 00:00:00
2	Bob	2021-06-10 00:00:00	2021-12-10 00:00:00
3	Charlie	2023-03-22 00:00:00	2023-09-22 00:00:00

Question 2: Use DATEDIFF() to find age in days from dob to today

Table 2: students

Student_id	name	dob
101	Maya	2005-08-10
102	Ethan	2004-12-01
103	Sienna	2006-03-15

```
SELECT
        student_id,
        name,
    DATEDIFF(day, dob, CURRENT_DATE()) AS age_in_days
FROM students;
```

Output Table

Student_id	name	age_in_days
101	Maya	7555
102	Ethan	7807
103	Sienna	7338

Question 3: Find how many days are left until each event using DATEDIFF().

Table 3: Events

event_id	event_name	event_date
1	Seminar	2024-06-15
2	Workshop	2025-09-01
3	Hackathon	2025-01-20

```
SELECT
    event_id,
    event_name,
DATEDIFF(day, CURRENT_DATE(), event_date) AS
days_remaining
FROM events;
```

Output Table

event_id	event_name	days_remaining
1	Seminar	-671
2	workshop	-228
3	Hackathon	-452

Question 4: Calculate the number of days between issue_date and due_date.

Table 4: Invoices

Invoice_id	Issue_date	Due_date
501	2025-03-10	2025-03-25
502	2025-04-01	2025-04-15
503	2025-04-10	2025-04-20

```
SELECT invoice_id,
       issue_date,
       due_date,
       DATEDIFF(days, issue_date, due_date AS days_between
FROM Invoices;
```

Output Table

Invoice_id	Issue_date	Due_date	Days between
501	2025-03-10 00:00:00	2025-03-25 00:00:00	15
502	2025-04-01 00:00:00	2025-04-15 00:00:00	14
503	2025-04-10 00:00:00	2025-04-20 00:00:00	10

Question 5: Format start_date as 'Month yyyy' using TO_CHAR()

Table 5: Courses

Course_id	name	start_date
201	SQL Basics	2025-05-01
202	Python	2025-06-01

```
SELECT
    Course_id,
    name,
  TO_CHAR(start_date, 'Month yyyy') AS
   Formatted_date
FROM Courses;
```

Output Table

Course_id	Name	Formatted-date
201	SQL Basics	May 2025
202	Python	June 2025

Question 6: Create full date from parts using DATE_FROM_PARTS()

Table 6: Memberships

member_id	start_year	start_month	start_day
1	2023	5	10
2	2023	11	25

```
SELECT member_id,
  DATEFROMPARTS(start_year, start_month, start_day) AS
      full_start_date
FROM Membership;
```

Output Table

Member_id	full_start_date.
1	2023-05-10 00:00:00
2	2022-11-25 00:00:00

Question 7: Extend each renewal_date by 1 Year using DATEADD

Table 7: Subscriptions

Sub_Id	Plan	Renewal_date
11	Basic	2025-01-01
12	Premium	2025-03-15

```
SELECT sub_id,
       Plan,
   DATEADD(Year, 1, renewal_date) AS extended_renewal_date
FROM subscriptions;
```

Output Table.

Sub_id	Plan	Extended_renewal_date
11	Basic	2026-01-01 00:00:00
12	Premium	2026-03-15 00:00:00

Questions8: Show current date and difference from order_date.

CURRENT_DATE and DATEDIFF().

Table 8: Orders

Order_id	order_date
1001	2025-04-15
1002	2025-04-10

```
SELECT order_id,
        Order_date, CURRENT_DATE() AS today_date, DATEDIFF(day,
        order_date, CURRENT_DATE()) AS days_since_order
FROM orders;
```

Output Table.

order_id	order_date	today_date	days_since order
1001	2025-04-15 00:00:00	2026-04-17 00:00:00	367
1002	2025-04-17 00:00:00	2026-04-17 00:00:00	372

Question 9: Extract the year from training_date using DATE_PART() OR EXTRACT()

Table 9: Trainings

training_id	topics	training_date
1	Safety	2025-01-10
2	Compliance	2025-02-20

```
SELECT
    training_id,
    topic,
  EXTRACT(YEAR FROM training_date) AS training_year
FROM Trainings;
```

Output Table

Training_id	Topic	training_year
1	Safety	2025
2	Compliance	2025

Question 10: Extract hour and minute from published_on.

Table 10: Blog_Posts

Post_id	title	Published_on
1	SQL Tips	2025-04-01 10:15:00
2	Data cleaning	2025-04-12 16:45:00

```
SELECT
        Post_id,
        title
    EXTRACT (HOUR FROM published_on) AS hour_published,
    EXTRACT (MINUTE FROM published_on) AS minutes_published
FROM Blog_Posts;
```

Output Table

Post_id	title	hour_published	minute_published
1	SQL Tips	10	15
2	Data Cleaning	16	45

Question 11: Calculate days left until license expiry using DATEDIFF() and today's date.

Table 11: Drivers

drive_id	license_expiry
301	2025-08-10
302	2023-12-31

```
SELECT
    driver_id,
    license_expiry
DATEDIFF(day, CURRENT_DATE(), License_expiry) AS day_left
FROM Drivers;
```

Output Table.

driver_id	license_expiry	days_left
301	2025-08-10 00:00:00	-250
302	2023-12-31 00:00:00	-838

Question 12: Display the current timestamp and calculate seconds since the message was sent.

Table 12: Messages

Message_id	Sent_timestamp
1	2025-04-19 09:32:45
2	2025-04-18 23:59:59

```
SELECT
        message_id,
        Sent_timestamp
    CURRENT_TIMESTAMP() AS current_timestamp,
    DATEDIFF(second, sent_timestamp,
    CURRENT_TIMESTAMP()) AS seconds_since_sent
FROM messages;
```

Output Table

Message_id	Sent_timestamp	current_timestamp	seconds_since_sent
1	2026-04-19 09:32:45	2026-04-17 04:03:07 117223	31343422
2	2026-04-18 23:59:59	2026-04-17 04:03:07 117223	31377768

Question 13: Add 15 days to return_date using DATEADD() to show restock_date.

Table 13: Returns

Return_id	Return_date
901	2025-04-05
902	2025-04-01

```
SELECT
    return_id,
    return_date,
    DATEADD(day, 15, return_date) AS restock_date

FROM Returns;
```

Output Table

Return_id	Return_date	restock_date
901	2025-04-05 00:00:00	2025-04-20 00:00:00
902	2025-04-01 00:00:00	2025-04-16 00:00:00

Question 14: Convert assigned_on to date using TO_DATE (if it's stored as string).

Table 14 Assignments

assign_id	assigned-on
1	2025-03-01
2	2025-03-05

```
SELECT
        assigned_id
    TO_DATE(assigned-on) AS assigned-on-date
FROM Assignments;
```

Output Table

assign_id	assigned_on_date
1	2025-03-01 00:00:00
2	2025-03-05 00:00:00

Question 15: Convert scheduled_time to formatted string like 'April 19, 2025 at 2:00 PM' using TO_CHAR().

Table: 15 Meetings

Meeting_id	Scheduled_time
1	2025-04-19 14:00:00
2	2025-04-19 09:30:00

```
SELECT
    meeting_id,
  TO CHAR(scheduled_time, 'Month DD, YYYY at HH:MI AM')
  AS formatted_meeting_time
  FROM Meetings;
```

Output Table

Meeting_id	Formatted_meeting_time
1	April 19, 2025 at 02:00 PM
2	April 19, 2025 at 09:30 AM